Pea diseases

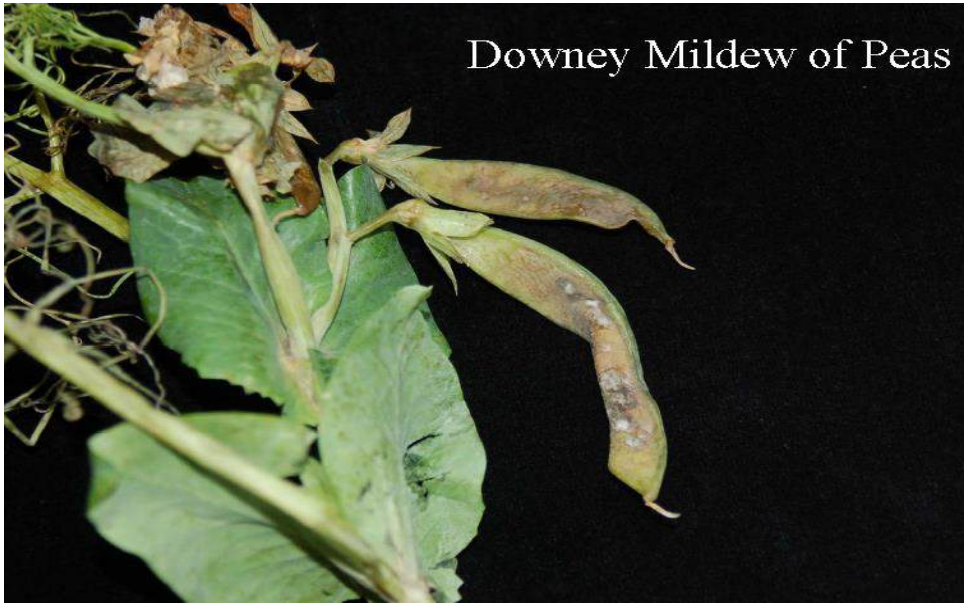
1. Pythium seed and root rot - *P. aphanidermatum*, *P. debaryanum*, *P. ultimum*
2. Downy mildew - *Peronospora pisi*
3. Powdery mildew - *Erysiphe pisi*
4. Rust - *Uromyces viciae fabae*, *U. pisi*
5. Fusarium wilt - *Fusarium oxysporum* f. sp. *pisi*
6. Anthracnose - *Colletotrichum pisi*
7. Ascochyta blight or foot rot or Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella pinodes*)
   Leaf and pod spot- *Ascochyta pisi* (PS: *Didymella pisi*)
8. Cercospora leaf spot - *Pseudocercospora cruenta*
9. Bacterial blight - *Pseudomonas syringae* pv. *syringae*
10. Mosaic - *Pea Seed Borne Mosaic Virus* (*PSbMV*), *Common pea mosaic virus* (*CPMV*)
11. Enation - *Pea Enation Mosaic Virus*

Pythium seed and root rot - *P.aphanidermatum, P. debaryanum, P. ultimum*

Pythium seed and root rot - *P.aphanidermatum, P. debaryanum, P. ultimum*

Pythium seed and root rot- Refer tomato

Downy mildew - *Peronospora pisi*



Infected pods

Downy mildew - *Peronospora pisi*

Sporulating systemic leaf infection.

Downy mildew - *Peronospora pisi*

External lesions on the pod

Infected pod showing internal signs of the pathogen along the dorsal sutures of the pods.

Downy mildew - *Peronospora pisi*

- Important disease of pea particularly under **high rainfall conditions and also at 22^0 C with low night temperatures.**

Symptoms on leaves

- **Local infection** is first on lower leaves and then spread to upper leaves.
- Noticeable at the time of **third and fourth leaf stage.**
- Scattered **yellow to brown circular to irregular** patches appear on the upper surface of the **leaves, stipules, petioles and some times on stem.**
- On the lower surface, white to **grayish violet downy growth** of the fungus is observed.

Downy mildew - *Peronospora pisi*

Symptoms contd…

- Downy growth consists of sporangiophores & sporangia.
- **Infected tissues soon turn brown and die.**
- **In case of systemic infection stunting and distortion of plants, killing the plant before flowering**

On Pods

- Can be seen when pods are flat and green.
- The patches on pods are **pale green, more or less elliptical** mostly **on sides along the dorsal sutures of the pods**.
- Pod lesions gradually becomes brown.
- **Seeds beneath the lesions are aborted** and under sized.
- **Fungal growth is** abundant on leaves, **rare on pods**

Downy mildew - *Peronospora pisi*

Management

- Destruction of infected plant debris and deep ploughing
- Use of healthy seed
- Three year or long crop rotation.
- Wider spacing
- Tolerant varieties *i.e*, Gastro, starcovert, starnain, Puget, cobra and mukta
- Seed treatment
- Spraying of fungicides carefully to reach the lower leaves.

Powdery mildew - *Erysiphe pisi*

Symptoms on the leaf and stipule of a field pea

Symptoms on the foliage and pod of a field pea

Powdery mildew - *Erysiphe pisi*

Powdery mildew resistant plant beside a badly infected plant

Powdery mildew - *Erysiphe pisi*

- Powdery mildew is one of the most common and serious diseases of pea which is prevalent throughout the world.
- In India this disease was first time reported by Butler (1918) from Dehradun.

Symptoms:

- Initial symptoms appear as minute discoloured patches with thin fine meshwork of white mycelium arising at many places on the **upper surface of lower leaves** and these white patches soon join together to form larger whitish floury areas.

Powdery mildew - *Erysiphe pisi*

Symptoms contd…

- In severe cases of disease development, both upper and lower surfaces of the leaf are infected.
- The conidia and conidiophores are produced profusely hence foliage looks as if dusted with flour and the **whole crop in the field appeared white from a distance.**
- With the progress of the disease, similar symptoms are also noticed on **tendrils, petioles and stems**

Powdery mildew - *Erysiphe pisi*

Symptoms contd..

- Pods are infected in all the developmental stages, but **immature pods are more prone** to rapid infection.

- The affected pods show white floury patches consisting of white powdery mass and these patches then turn light **brown and finally dark brown in colour** and later become necrotic.

- Such **infected pods remain small, shrivelled with less weight** and unmarketable/fetch minimum price

Powdery mildew - *Erysiphe pisi*

Pathogen:

- The fungus is ectoparasitic, spreading on the surface of the host and sending haustoria into the epidermal cells to draw out nourishment.
- The fungus hyphae are hyaline and septate and conidiophores arise from it vertically bearing conidia either singly or in short chains
- Conidia are ellipsoid to ovoid, mature conidia fall off and get dispersed by wind
- Later in the season, **cleistothecia** (sexual stage) appear as dark coloured, round, minute bodies scattered in the mycelial web.

Powdery mildew - *Erysiphe pisi*

Pathogen contd…

- The peridium is composed of distinct polygonal cells and is provided with a number of myceloid appendages.
- Each **cleistothecium contains usually 2-8 asci** and these asci originate from a single point in the fruiting body, appearing in a fanlike manner.
- They are **ovate, nearly sessile** and 46-72 x 30-45 µ in size.
- The asci contain 2-8 ascospores**, which are elliptical, hyaline and unicellular.**

Powdery mildew - *Erysiphe pisi*

Disease cycle and epidemiology:

- **Dormant mycelium or cleistothecia in plant debris** or on the standing crop as a source of primary inoculum

- There are reports of pathogen survival in seed as dormant mycelium but the **seed borne nature of the disease was not confirmed.**

- However, in other areas, the pathogen overwinters in **conidial stage on different collateral hosts** of the pathogen like *Pisum arvense*, *Lupinus*, *Medicago*, *Lathyrus aphaca*, *L. culinaris*, *Medicago hispida*, *Melilotus inica*, *Vicia sativa*, *V. hirsute* and *Aeschynomene indica*.

Powdery mildew - *Erysiphe pisi*

Disease cycle and epidemiology:

- Ascospores or conidia germinate by producing a germ tube which produces an appresorium on its tip and continues growing ectophytically.
- Each conidia has 75 % moisture and germinate easily under dry conditions, so it is a dry weather fungus.
- Conidiophores bearing conidia are produced by the growing mycelium.
- Conidia are passively dispersed and cause secondary spread of the disease.

Powdery mildew - *Erysiphe pisi*

Disease cycle and epidemiology contd..

- The disease is more prevalent in **dry weather and moderate temperatures.**
- A temperature range of **10-30°C** is favourable for conidial germination with optimum being at 20°C.
- **A fairly dry soil** and **heavy application of nitrogenous** fertilizers increase the disease incidence.

Powdery mildew - *Erysiphe pisi*

Management:

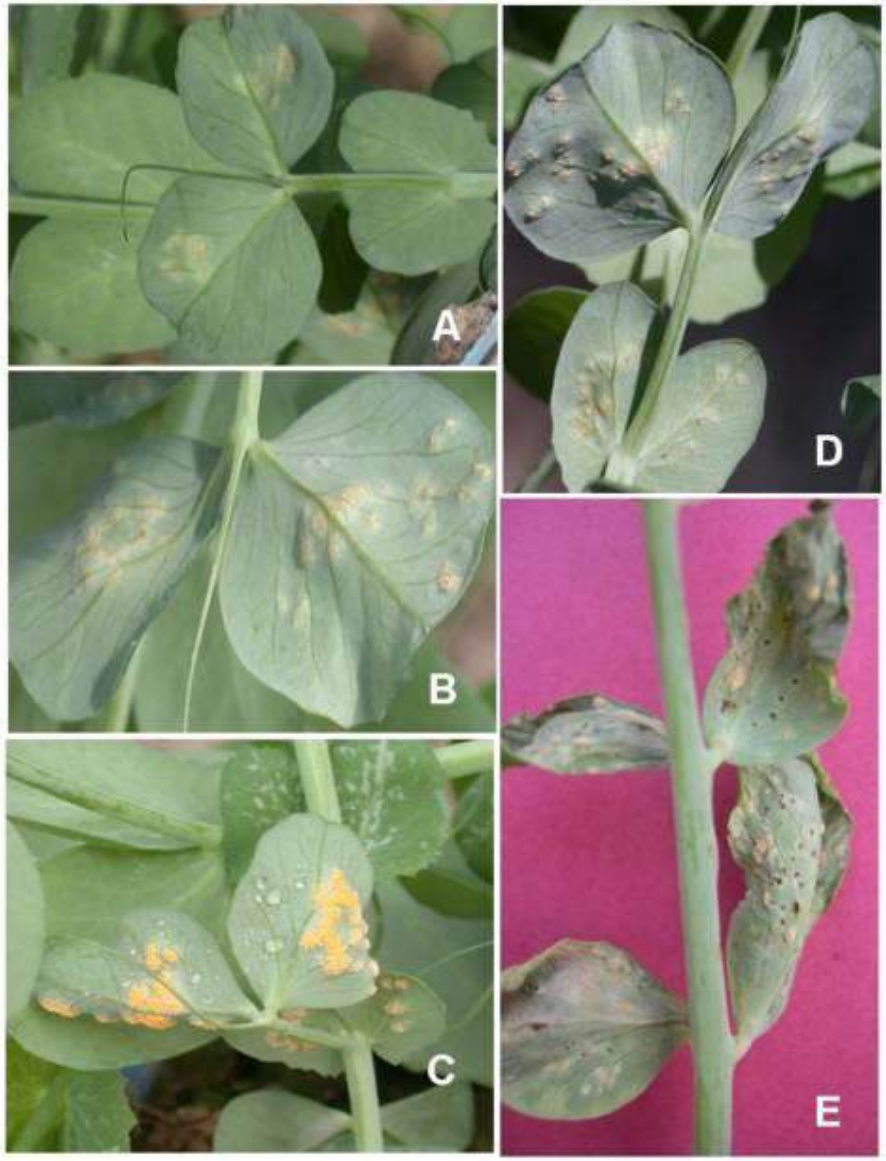
- Collect and destroy the plant debris.
- Remove all the collateral hosts from in and around the field.
- Early sown crops (Late September to early October) escape the disease.
- A large number of cvs. /lines possessing resistance can be used for sowing or in breeding programme for developing resistant varieties (Solan nirog, Rachna etc.).

Powdery mildew - *Erysiphe pisi*

Management contd…

- Host plant resistance : **Pant P-9, Rachana, DMR 11**
- Spraying of plant extracts (garlic) and oils (neem)
- Use of biocontrol agent like *Ampelomyces quisqualis* serve as a best alternative
- With the initiation of flowering, spray the crop with wettable sulphur (0.20 %) dinocap (0.05 %) or carbendazim (0.05%) or hexaconazole (0.05%) or triadimefon (0.1%) or fenarimol (0.04%) or difenoconazole (0.03%) and repeat at 10-14 days interval.

Rust - *Uromyces viciae fabae, U. pisi*



A)Symptoms development of *U. viciae-fabae* on pea after infection
B) Aecia development after inoculation by Uredia
C) Acedial pustules on the abaxial side of the leaf
D) Small uredial pustules
E) Acedial and Uredial pustules.

Rust - *Uromyces viciae fabae, U. pisi*

(A) Uredial
(**B**) Telial pustules on stems of pea

Rust - *Uromyces viciae fabae, U. pisi*

- Pea rust is an important disease, **which occurs quite frequently in pea growing areas** in North and North western parts of India.

- Two rusts have been reported on peas caused by ***Uromyces viciae - fabae* (worldwide occurrence)** and *Uromyces pisi*

Symptoms:

- The symptoms appear on all above ground parts of the plant, initially **more pustules on the lower surface of leaves**, **less abundant on pods, while sparingly on stem**

Rust - *Uromyces viciae fabae, U. pisi*

Symptoms contd…

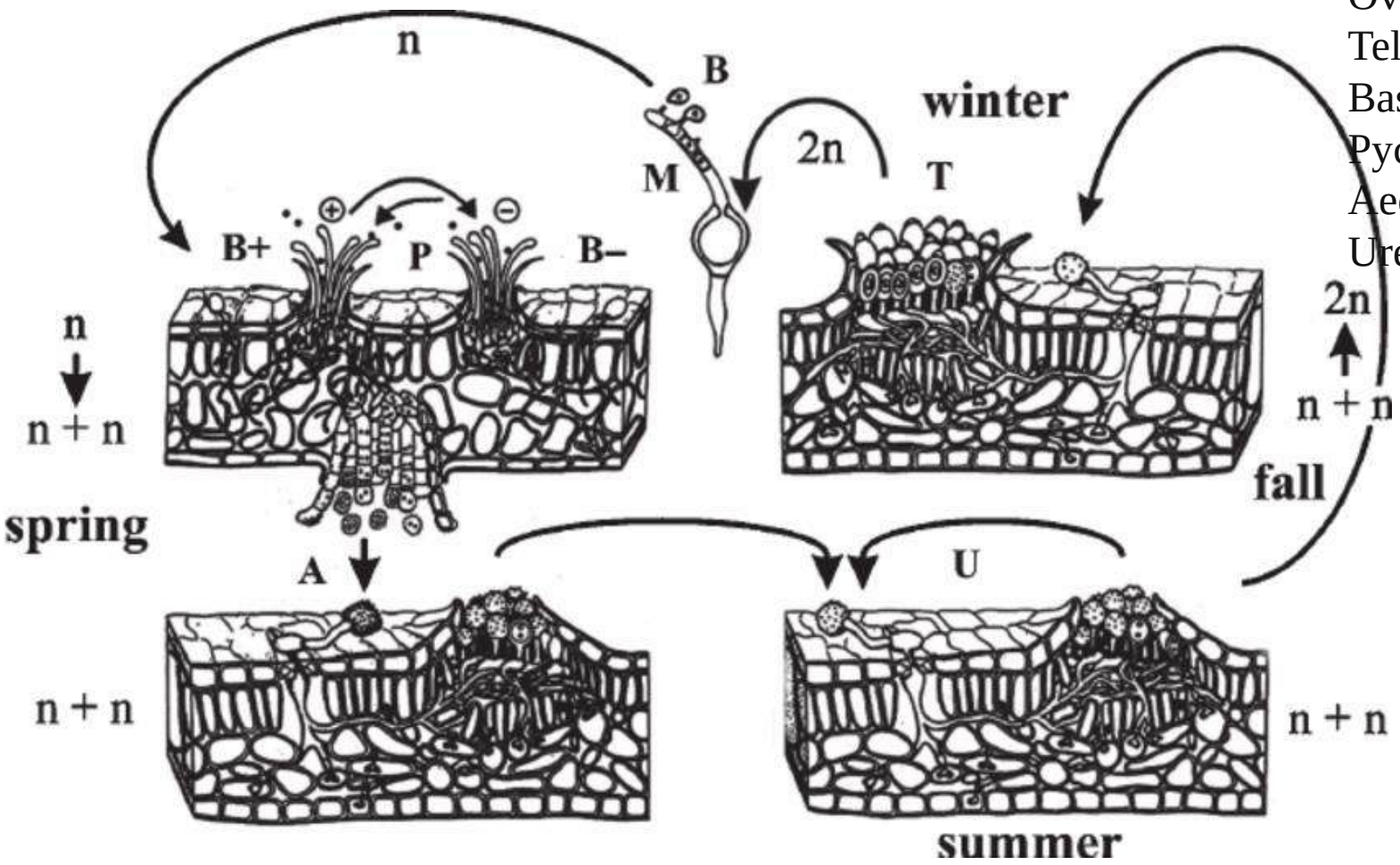
- Infection is evident first as **minute, raised pustules, which later become distinct, yellowish, circular sori**
- The **yellow spots have aecia** and persist for longer time.
- The uredo pustules develop on both surfaces of the leaves and on other parts and give **light brown powdery appearance** to the plants in case of severe infections.
- Late in the season, **teleuto pustules** also develop on the leaves but **most commonly on stems and petioles**.
- They are **dark brown or almost black in colour.**

Rust - *Uromyces viciae fabae, U. pisi*

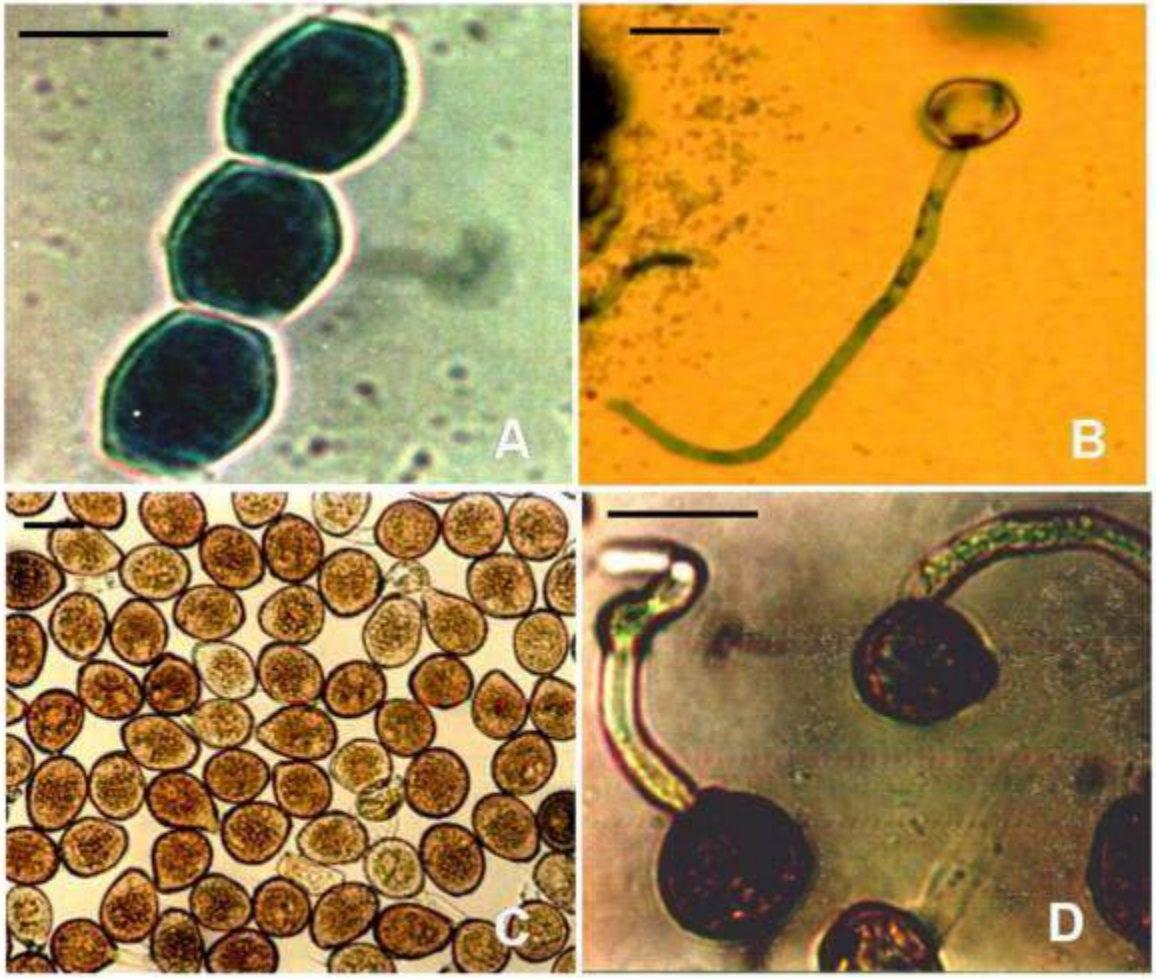
Pathogen:

- On peas, the pycnia occur in small groups associated with the aecia.
- The aecia are cup shaped and 0.3 to 0.4 mm in dia.
- The peridium is short and whitish.
- The aeciospores are round to angular or elliptical with hyaline wall and the wall of these spores is verucose.

Uromyces viciae fabae - life cycle



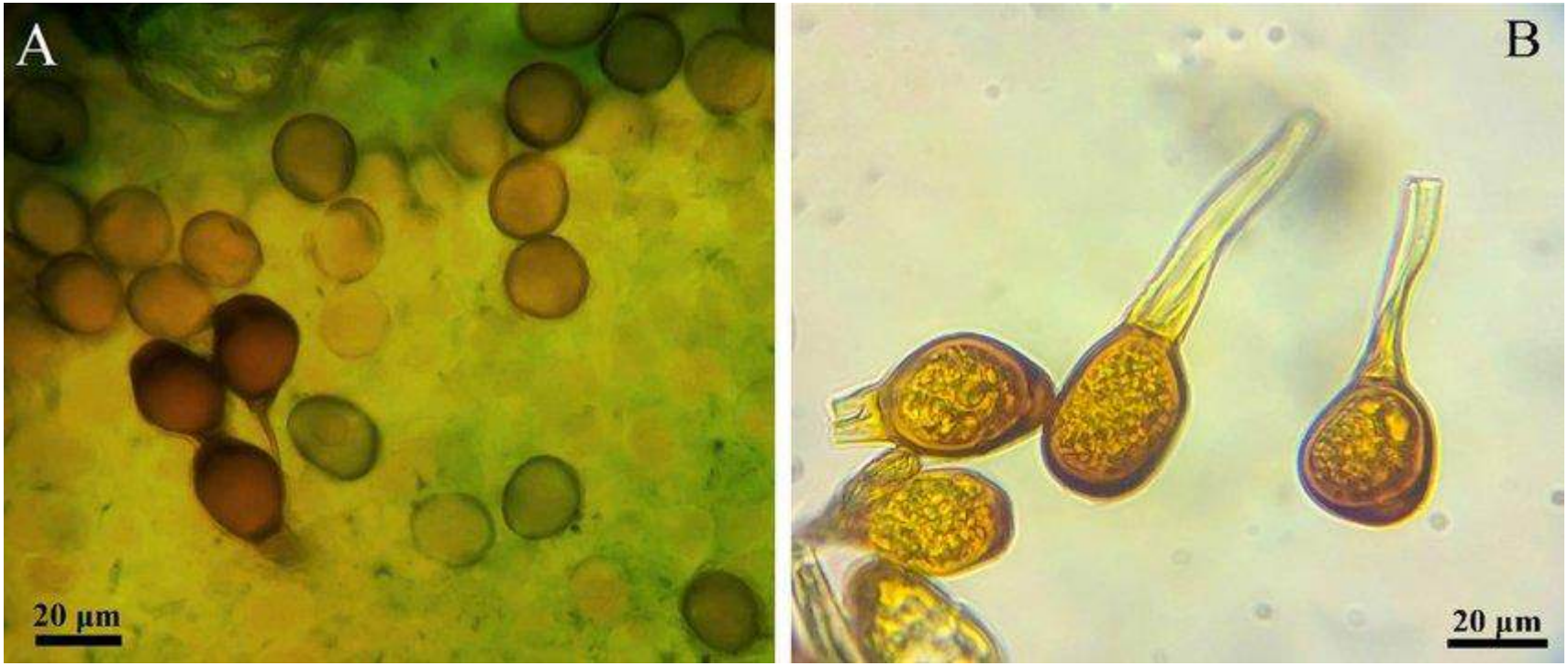
Overwintering
Teliospores (T)
Basidiospores (B)
Pycniospores (P) a
Aeciospores (A)
Uredospores (U)



Aeciospores ,**B**) Germinating aeciospores, aecial infections reach the mesophyll and spongy tissues to form the aecial cups

C) Urediospores (**D**) Germinating urediospores, uredial is mostly confined to the epidermal cells and a few layers of mesophyll cells

Uromyces viciae fabae - life cycle



Uredospores (A) and teliospores (B)

Rust - *Uromyces viciae fabae, U. pisi*

Pathogen contd…

- The urediniospores are round to ovate, light brown, echinulate, with 3-4 equatorial germ pores.
- The telia are **dark brown to black occur** in the same sorus as the uredia
- The teliospores are **sub-globose, ovate or elliptic with rounded or flattened apex,** which is considerably thickened and appears **papillate,** smooth and are brown in colour.

Rust - *Uromyces viciae fabae, U. pisi*

Disease cycle and epidemiology:

- **An autoecious rust** - all its spore stages on the same host.
- The rust is **seed borne in broad beans** but in peas and lentil, it is **mainly survive as teliospores in crop debris**.
- **Aecio and uredospores can not tolerate summer temp. in plains of India, hence, the rust appears to survive on weed hosts** belonging to *Lathyrus*, *Vicia* etc. and the spores are wind blown to the main crop.
- The teliospores on germination produce a promycelium on which sporidia are borne.
- The sporidia germinate and germ tube infects the host, **produce pycnia and finally aecial cups are produced.**

Rust - *Uromyces viciae fabae, U. pisi*

Disease cycle and epidemiology contd…

- **Uredia occur late in the season and are not of much importance in secondary spread**.
- Later in the season, teliospores are produced which help in overwintering of the pathogen.
- Prevalence of low temperature (17-22° C) coupled with high humidity, dew or frequent light rain showers are favourable for **initiation of the disease by aeciospores**.
- Rust severity and pustules/plant increase progressively with an increase in the duration of leaf wetness up to 24 h, at optimum temperature of 20°C.

Rust - *Uromyces viciae fabae, U. pisi*

Disease cycle and epidemiology contd…

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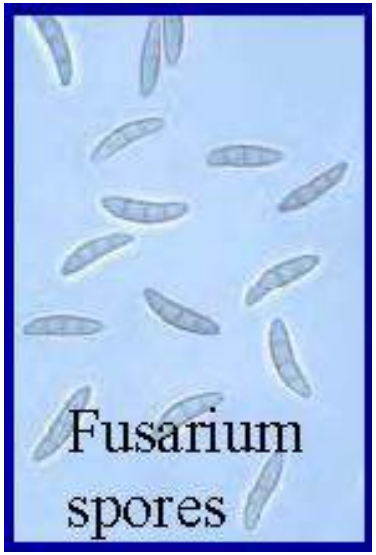
Rust - *Uromyces viciae fabae, U. pisi*

Management:

- Collect and destroy infected plant debris.
- Follow long crop rotations avoiding **broad beans, *Vicia* and *Lathyrus* in rotation.**
- In disease prone areas early sowing can be taken up
- With the initiation of the disease, spray the crop with mancozeb (0.25%) or hexaconazole (0.1%) or triadimefon (0.1%) or difenoconazole (0.05%) and repeat at 10 to 14 days interval depending on the severity of the disease.

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

Fusarium rot - *Fusarium oxysporum* f.sp. *pisi*



Fusarium rot - *Fusarium oxysporum* f.sp. *pisi*

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

- This is an important disease prevalent in those areas where **pea is cultivated in autumn.**
- In northern plains of India, wilt is a serious problem in the early sown (September planted) pea crop.

Symptoms: **true wilt and near wilt**

True wilt:

- Initially, the lower leaves look pale yellow and droop downwards
- With the progress of the disease, similar symptoms appear on the upper leaves also and the affected plants give a stunted look.

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

Symptoms contd…

- Slowly the upper plant parts loose their turgidity and the whole plants topple down.
- Stems are shriveled and some of the root lets are also injured.
- Plants wilt or die due to **destruction of feeder roots**
- Affected plants die **more rapidly in dry soil than wet soil**
- **If infects young plants, die without pod formation**
- If late in the season a few flattened and poorly filled pods

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

- **Yellow to orange brown** discolouration of vascular bundles of upper tap root extending up to the stem
- Disease usually **in circular to pattern in field**

Near wilt:

- Varieties **resistant to true wilt are affected**
- **Losses are less than true wilt** as it occurs late in the season some times when the crop is nearly mature
- When the older plants are affected **a black canker develops on the underground stem** which extends upward and downward

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

Near wilt contd…

- Symptoms are also similar to those described above except for their slow development.
- In this case, the **vascular discolouration of the roots is typically brick red and usually extends up to the growing tip.**
- **Cortical decay in lower parts of the stem and upper tap root.**
- Disease occurrence is circular pattern in the field

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

Disease cycle and epidemiology:

- The pathogen is soil borne and can survive in the soil for **more than 10 years as chlamydospores**.
- It **can be carried on seed coat**
- The fungus causes infection of the fibrous roots or epicotyl region, grows both inter and intracellularly in the cortex and ultimately concentrates in the xylem vessels.
- After death of the plant, the fungus continues to grow and sporulate on the stem cortex, resulting in the production of soil borne inoculum.

Fusarium wilt - *Fusarium oxysporum* f.sp. *pisi*

Disease cycle and epidemiology contd….

- The disease progresses fast at 21° C while near wilt requires 24-28° C.
- **True wilt is not serious in very wet soils, while wet soils favour near wilt.**
- Neutral to **alkaline soils favour true wilt** while the acidic soils retard it.

Fusarium wilt - *Fusarium oxysporum* Schlech. ex. Fr. f.sp. *pisi*

Management:

- Collect and destroy the infected plant debris.
- Follow long crop rotation in the infested fields.
- Kala nagini variety moderately resistant
- Use healthy seed and seed treatment carbendazim (0.2%) or Seed treatment with Trichoderma spp.
- Soil pH - 7.4
- **Balanced application of nitrogen and phosphorus fertilizers**

Fusarium wilt - *Fusarium oxysporum* Schlech. ex. Fr. f.sp. *pisi*

Management contd…

- Late sowings 1st and 15 th November when the soil temp is low then less wilt development
- **Soil flooded with water develops less severe wilt** than the non flooded soil
- Biological control with ***Pythium ultimum* and *Gliocladium roseum***

Anthracnose - *Colletotrichum pisi*

Anthracnose - *Colletotrichum pisi*

Symptoms

- Lesions develop on leaves, stems and pods

On leaves

- **Leaf and stipule**, lesions are oval, 2-8 mm in diameter, and have **brown margins with gray-tan centers.**
- **On stems, spots are elliptical** with grey centre
- **Pod lesions are round** and tan gray-and sunken, with reddish- brown borders.
- These are most dramatic when they form on immature pods, causing them to develop abnormally and **show brown discoloration**

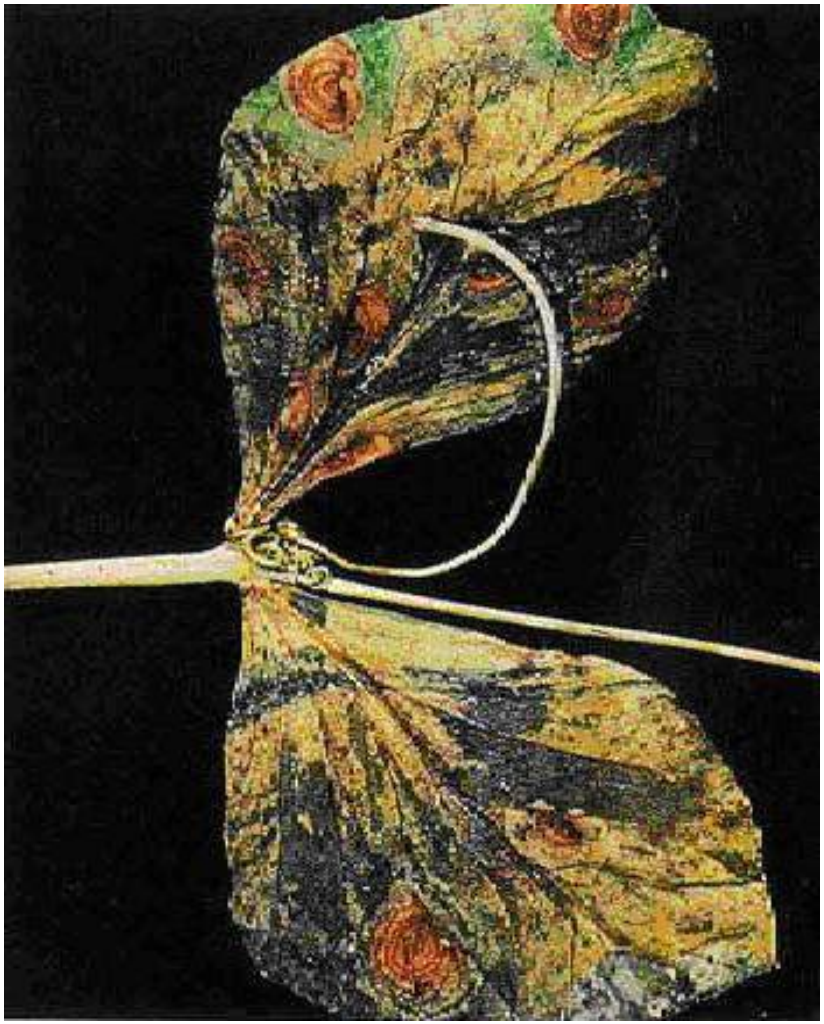
Anthracnose - *Colletotrichum pisi*

Management

Refer Bean anthracnose

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)



Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

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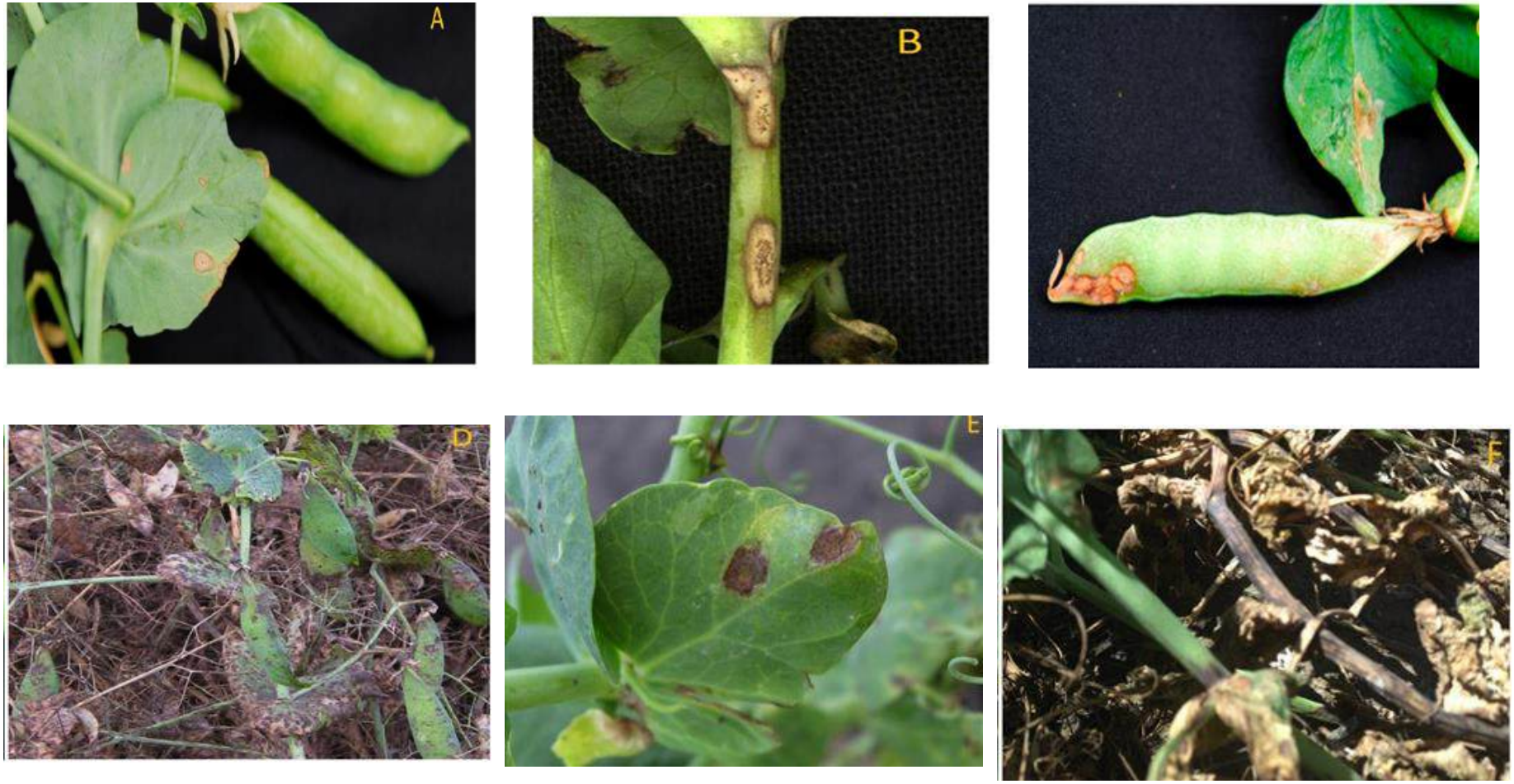


A. *Ascochyta pisi* lesions on leaves B. *A. pisi* lesions on stem C. *A. pisi* pod infection
D. *P. pinodes* lesions on plants E. *Peyronellaea pinodes* symptoms on leaves
F. *P. pinodes* girdling of stem

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

- Ascochyta blight commonly known as **black spot**, is one of the most severe diseases of field peas
- **A complex of fungal pathogens and** commonly referred to the **Ascochyta complex** *A. pinodes*, *A. pinodella*, *A. pisi* and ***Phoma medicaginis* var. *pinodella, Phoma koolunga***
- This blight complex causes a range of different symptoms, including Ascochyta blight, foot rot, black stem and leaf and pod spot.
- **Seed quality may also be reduced through seed discoloration or retardation of seed development.**

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

- ***A. pinodes* can infect seedlings and all aerial parts of pea plants,** causing necrotic leaf spots, stem lesions, shrinkage and dark-brown discoloration of seeds, blackening of the base of the stem, and **foot rot in seedlings**.

- The disease symptoms caused by *P. pinodella* are similar to those observed with *A. pinodes*.

- However, *P. pinodella* infection can result in more severe foot rot symptoms that can extend below ground, while causing less damage to the leaves, stems and pods.

- ***A. pisi* usually does not attack the base of pea plants or cause foot rot,** causes slightly sunken, circular, tan-colored lesions with dark brown margins that occur on the leaves, pods and stems.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

- *P. koolunga* presents disease symptoms on pea seedlings that are indistinguishable from those caused by *M. pinodes*

- Recently, *Phoma herbarum* and *P. glomerata* have also been shown to be associated with the Ascochyta blight complex on field peas in Australia, causing typical dark brown lesions and chlorotic halos on pea leaves.

- **All related pathogens are seed-borne pathogens** that can also survive on infected pea debris.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

- **The morphologically distinct species of Ascochyta** with different types of symptoms including **leaf and pod spot, blight and fruit rot.**

Symptoms: Ascochyta or Mycosphaerella blight:

- **On foliage-** Initial spots **brown to purplish, irregular areas later become circular and somewhat zonated.**

- **On pods** - Small, **brown to purplish irregular spots, later** enlarge to cover large area and coalescence of lesions - blotched

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Symptoms contd..

- **Developing seeds may be deformed**, remain small in size and are sometimes stained.
- **Black to purplish streaks** appears on stem.
- In lesions, **pycnidia appear in characteristic ring pattern**.

Ascochyta foot rot:

- **The symptoms are similar to those described for blight but in foot rot lesions predominate.**
- The dark **brown lesions on the stem start soon after germination** below the soil line and gradually extend down to roots and up the stem.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Ascochyta leaf and pod spot *A. pisi*

- Symptoms on **leaves and pods appear as definite, circular, sunken, tan to brown spots while on stems and petioles elongated spots**
- **A dark brown margin usually surrounds** the tan spot and pycnidia may form on these spots also.
- Underlying seeds in the pods may also be affected and remain shriveled.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Ascochyta

- **Perithecia** are globose having beaked ostioles and contain bitunicate, cylindrical, clavate asci.
- **Ascus** contains hyaline, bicelled eight ascospores
- The fungus colonies are light to dark gray in colour and often have concentric rings.
- The conidia have large number of globules.
- The spore ooze collectively appears buff coloured.

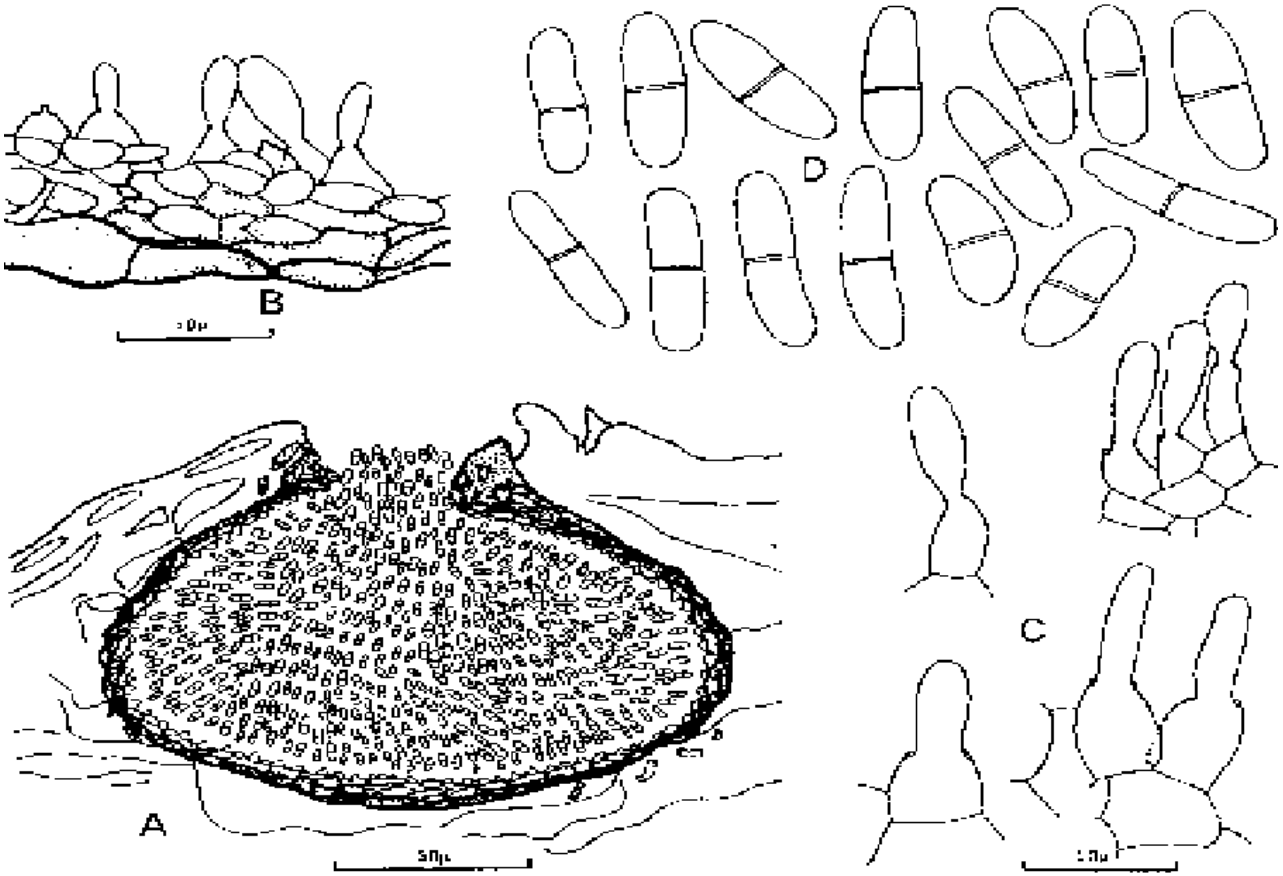
Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

- ***A. pinodella* -** This **is not known to produce perfect stage,** how ever, fungus produces numerous **chlamydospores singly or in chains.**

- ***A. pinodella* -** Colonies are dark gray and turn black at maturity while in ***A. pisi*** light in colour and exude spore mass which is carrot red in colour.

- ***A. pinodella*** – Pycnidiospores are hyaline, single or bicelled, while ***A. pisi*** Conidia are longer and thin with slight constriction at septation.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)



A. Immersed pycnidium B. Colony of conidiophore on substratum C. Conidiophore D. Conidia

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella pinodes*) Leaf & pod spot - *A. pisi* (*PS: Didymella pisi*)

Disease cycle and epidemiology:

- Different Ascochyta species **perpetuates either in the seed or in diseased crop debris.**
- In crop debris, these species **survive as perithecia, sclerotia or chlamydospores.**
- The pathogens may also survive through **collateral hosts** which are *Lathyrus* sp. and *Vicia* spp., for *A. pisi*; *Phaseolus* and *Lathyrus* for *A. pinodes* and *Vicia trifolium* and other legumes for *Phoma medicaginis* var. *pinodella*).

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Disease cycle and epidemiology contd…

- **Cool weather coupled with presence of sufficient moisture on the leaves** is the most favourable conditions for disease development.
- The disease attains serious proportions **during prolonged rainy periods.**
- Atmospheric humidity >90 % helps in increasing the disease incidence while **no infection occurs below 80 % RH.**

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Management:

- Use disease free seeds.
- Disease free seeds can only be produced if the crop is raised in low rainfall areas.
- The soil borne inoculum can be reduced by following long crop rotations and by destruction of the crop refuse by burning either in the field or after threshing.
- Use resistant sources, if available.

Ascochyta blight / foot rot / Mycosphaerella blight - *Ascochyta pinodes* (PS:*Mycosphaerella/ Didymella pinodes*) Leaf & pod spot - *A. pisi* (*PS: D. pisi*)

Management contd…

- Seed dressing with microbial cultures of *Trichoderma koningii* and *Gliocladium roseum* (4 g/kg seed) was also found effective in protecting the crop against *A. pisi*.
- Treat the seed with fungicides namely carbendazim (0.2%), benomyl (0.1%), thiabendazole (0.1%) + thiram (0.3%) and captan (0.3%).
- With the initiation of the disease spray the crop with fungicides like chlorothalonil (0.2%), carbendazim (0.05%) or combination of both and repeat at 10 days interval.

Cercospora leaf spot - *Cercospora cruenta*

Cercospora leaf spot - *Pseudocercospora cruenta.*

Symptoms

- With Cercospora leafspot a dark, moldy growth develops on the lower leaf surface opposite the spot.
- Leafspot diseases are most serious during periods of prolonged moist weather and on late summer or early fall plantings.
- Severe leaf spotting results in defoliation with subsequent yield reductions.

Cercospora leaf spot - *Pseudocercospora cruenta.*

Management

- Use disease-free seed for sowing.
- Removal of debris after harvest.
- Crop rotation for two to three years.
- There are no resistant varieties
- Spraying of recommended chemicals

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

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Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Water soaked lesion spreading into the leaf
from the base caused by bacterial blight.

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

- It is an important disease of pea in those areas where cool and moist climate is prevalent during the growing season.

Symptoms:

- **All aerial parts of the plants** are infected by the disease.
- Symptoms **on leaflets and stipules at the point of joint on the midrib and stem** respectively as small water soaked lesions usually developing into large irregular areas
- Lesions vary in colour, initially being **dark green and water soaked,** but generally becoming water soaked at the edge and **a lighter shade of brown at centre.**
- Under conditions of high moisture, **chocolate brown linear streaks are observed on the stem and petiole.**

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Symptoms:

- The **flower buds are** often killed before opening
- Infection from sepals spreads to peduncle and to pods killing the flowers
- Later on, **the whole stem turns chocolate brown, shriveled** leading to the death of the plants.
- Due to infection, **immature pods become chocolate brown, thin, twisted and shriveled**, lesions are bigger on older pods.
- Seeds developing in the pods are also discoloured and shriveled

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Pathogen:

- The disease is caused by bacterium *Pseudomonas syringae* pv. *syringae* van Hall and *P. syringae* pv. *pisi* (Sackett) Young et al.

- The bacterium is **gram -ve, non-spore forming** and motile rods with one or more polar flagella.

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Disease cycle and epidemiology:

- The pathogen **overwinters in the infected seed and plant debris.**
- Alternate hosts- filed bean, soyabean, common bean, sweat pea and Vigna spp
- The seed carries the **bacterium both externally and internally .**
- It colonizes the intercellular and intracellular spaces of the seed coat but **does not penetrate the embryo or cotyledons.**
- Even a **very low level of seed infection can cause economic loss,** since the disease can spread fast from primary infection foci.

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Disease cycle and epidemiology contd…

- The pathogen is carried in irrigation water, splashed by rain or blown in wet winds to other plants and **field infection usually occurs through stomata and wounds**.
- Cool, moist weather favours the disease while warm and dry weather retards it.
- The extent of **disease spread depends upon the frequency of rainy periods.**
- If soil is very wet at the time of emergence, it favours **blight transmission from the infected seed** to the plant.

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Management:

- Collect and destroy the infected plant debris.
- Follow at least three years crop rotation with non leguminous crops.
- Resistant varieties., pitica galbens, minunea americiii, ICAR 53/54 and victoria baragan
- Timely sowing -15 th November
- Give pre-sowing seed dip treatment in Streptocycline (150 ppm) solution for 90 minutes.
- Surface sterilization of seed with sodium hypochlorite 1%

Bacterial blight - *Pseudomonas syringae* pv. *syringae*

Management contd….

- Use disease free seed and hot water treatment **at 60°C/ 15min** though it reduces seed germination by 20-45%
- Restricted irrigation and providing proper drainage
- Avoiding sprinkler irrigation
- Wider spacing
- With the initiation of the disease, spray the crop with Streptocycline (100 ppm) and repeat at 7 days interval.

Pea seed borne mosaic - *Pea Seed Borne Mosaic Virus* (PSbMV)

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Pea seed borne mosaic - *Pea Seed Borne Mosaic Virus* (PSbMV)

Symptoms:

- Symptoms on susceptible varieties leaf roll, mosaic, leaf and stem malformation and stunting
- **Chlorotic pattern on the leaves followed by narrowing of leaves** coupled with downward rolling
- In the later stages**, apical malformation resetting and tendril curling**.
- **Symptomless infection common in peas**
- This is followed by **vein clearing** and production of **malformed flowers.**
- The size of **pods is reduced** which get distorted depending upon disease intensity and produce **infected and** shriveled **seeds with cracks on seed coat**

Pea seed borne mosaic - *Pea Seed Borne Mosaic Virus* (PSbMV)

Pathogen:

- The disease is caused by *Pea Seed Borne Mosaic Virus* which belongs to poty virus (PSbMV) group.
- The virus particles are filamentous and flexuous rods 770 x 12 nm in size and non- enveloped.
- The genome consists of single stranded RNA.

Pea seed borne mosaic - *Pea Seed Borne Mosaic Virus* (PSbMV)

Disease cycle and epidemiology:

- The virus is transmitted mechanically, or by seed or by aphid vectors either semi- persistently or non-persistently.
- The natural vectors are *Acyrthosiphon pisum*, *Aphis craccivora* and *A. fabae*.
- The virus has wide host range 47 plants belonging to 12 familiesbut significant in hosts like pea, lentil and broad bean with regard to economic importance, dissemination through seed.

Pea seed borne mosaic - *Pea Seed Borne Mosaic Virus* (PSbMV)

Management:

- Use resistant pea cultivars.
- Seed certification reduces the disease incidence
- Since seed borne viruses create inoculum source within field
- Use reflective mulches to reduce the incidence of this virus.
- Spray the crop with insecticides like Malathion (0.1%) or Metasystox (0.1%).

Mosaic- Common pea mosaic Virus - (CPMV)

Symptoms:

- **Not seed borne**
- Plants look pale, stunted and weak
- On stipules conspicuous mosaic mottling
- On fully developed young leaves and stipules mosaic mottling symptoms
- Older leaves with irregular white patches
- Light brown discoloured spots on petioles, stem and tendrils
- Plants flower late, produce smaller and fewer flowers and pods with less number of seed

Mosaic- Common pea mosaic Virus - (CPMV)

Transmission by aphid – ***Aphis craccivora***

Management

- Growing resistant varieties
- Vector control

Enation - *Pea Enation Mosaic Virus*

Enation - *Pea Enation Mosaic Virus*

Enation - *Pea Enation Mosaic Virus* (PEMV)

- This virus infects **pigeon pea, soya bean, chick pea, broad bean.**

Symptoms- on leaves

- Identified by **small outgrowth (enations) on the lower surface of the leaves**
- **First on younger leaves and stipules** as light green & dark green mottling.
- **Deformed leaves** on the branches.
- Terminal growth of the plant is checked with **shortening of internodes.**
- Tips of the plants looks **like rosette**.

Enation - *Pea Enation Mosaic Virus*

Symptoms -on pods

- **Enations are also found on pods and usually no pod formation only with small yellow seeds**.
- **Pods formed late in the season are roughed**, ridged and so **badly deformed** that they do not look like pods.

Enation - *Pea Enation Mosaic Virus*

The virus consists of two plus-sense RNAs

Spread by vector - Aphids

***Acyrthosiphon pisum* or** *Macrosiphum pisi*, *M.solanifolii*.

Management

- Growing resistant varieties **franklin and lifter**
- Uprooting and destructing the infected plants.
- Vector control.